CryNetV2: Hybrid Deep Learning Framework for Infant State Recognition

*A Three-Phase Dual-Stream Cross-Attention Architecture for Cry Classification

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Abstract—This paper presents CryNetV2, an advanced hybrid deep learning framework developed to classify infant cry audio into eight distinct physiological and emotional states. The research progresses through a rigorous three-phase methodology. Phase I establishes a classical Machine Learning baseline (Support Vector Machines) yielding a Macro F1-Score of 0.26. Phase II introduces a neural approach using a CNN-BiLSTM-Attention architecture, increasing the F1-Score to 0.354. Phase III represents the culmination of this work with a novel Dual-Stream Cross-Attention architecture that optimally fuses standard Mel-Spectrograms with adaptive Discrete Stockwell Transform (DST) features. Combined with a Meta-Ensemble fusion strategy, CryNetV2 achieves a Macro F1-Score of 0.382 on an extremely imbalanced dataset of 2,108 samples. The entire system is production-ready, encapsulated within Docker containers, provisioned on AWS ECS Fargate via Terraform Infrastructure-as-Code, and governed by a fully automated GitHub Actions CI/CD pipeline.

Index Terms—Infant cry analysis, CNN-BiLSTM, cross-attention, Stockwell transform, Docker, Terraform, ECS Fargate, healthcare AI, CI/CD

I. Introduction

Infant crying is a complex, non-stationary acoustic signal and serves as the primary pre-linguistic communication mechanism encoding critical information about an infant's well-being. Different cry types—such as hunger, pain, fatigue, and general discomfort—carry distinct acoustic signatures that can be systematically classified using machine learning methodologies [1]. The automation of this classification process holds significant potential to assist caregivers, particularly in Neonatal Intensive Care Units (NICUs) and domestic healthcare environments, by enabling rapid response to infant needs.

Traditional acoustic analysis has heavily relied on handcrafted feature sets, such as Mel-Frequency Cepstral Coefficients (MFCCs). However, these conventional techniques frequently fall short of capturing the intricate non-linear dynamics and temporal dependencies inherent in infant cry sequences. In this research, we propose a progressive three-phase framework designed to overcome these limitations and directly address the severe class imbalances typical of real-world clinical datasets.

Phase I establishes a highly interpretable baseline employing classical ML algorithms (SVMs and Random Forests) acting upon handcrafted feature vectors. While interpretable, this approach lacks the representational power needed for complex cry patterns. Phase II introduces CryNet, a fully deep learning-based architecture combining Convolutional Neural Networks (CNNs) for spatial extraction, Bidirectional LSTMs for sequential modelling, and Self-Attention mechanisms.

Building upon these foundations, Phase III introduces CryNetV2, incorporating three major innovations: (1) A Discrete Stockwell Transform (DST) stream that delivers adaptive time-frequency resolution [2]; (2) Cross-Modal Attention (CMA) for the bidirectional fusion of Mel and DST streams [3]; and (3) A Weighted Meta-Ensemble that harmonises classical ML precision with deep learning recall capabilities.

II. Dataset Characteristics

The dataset under investigation comprises 2,108 infant cry recordings, evenly split between 1,054 cleaned original samples and 1,054 noise-augmented variants. The dataset is characterised by a severe 36:1 class imbalance ratio, making standard accuracy metrics highly misleading and necessitating the use of the Macro F1-Score for robust evaluation.

Table I delineates the exact class distribution following augmentation. The Hungry class heavily dominates the dataset (36.2%), whereas critical but rare states like Lonely (1.0%) and Scared (2.6%) are drastically under-represented.

The distinct spectral signatures of these classes can be observed in Fig. 1. These log-mel spectrogram visualisations illustrate how energy distribution across frequency bands varies significantly depending on the infant's physiological state. For instance, pain-induced cries typically exhibit higher fundamental frequencies and more chaotic

TABLE I
Dataset Class Distribution (After Augmentation)

Cry State	Samples	Proportion (%)
Belly Pain	254	12.0
Burping	236	11.2
Cold/Hot	228	10.8
Discomfort	274	13.0
Hungry	764	36.2
Lonely	22	1.0
Scared	54	2.6
Tired	276	13.1

high-frequency harmonic structures compared to standard hunger cries.

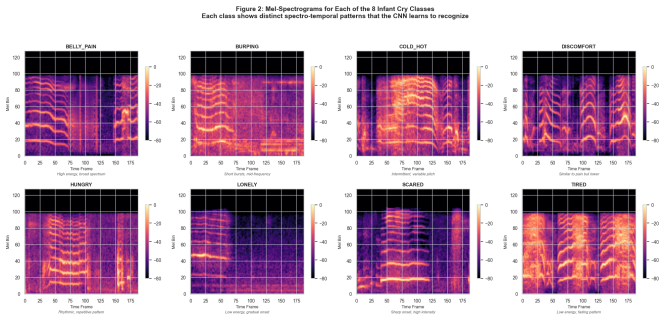

Fig. 1. Per-class log-mel spectrogram visualisations highlighting the distinct spectral energy patterns across the eight infant cry states. High energy concentrations in higher frequency bands are indicative of distress.

III. Methodology

Our methodology is structured into an extensive audio preprocessing pipeline followed by three distinct phases of architectural evolution.

A. Audio Preprocessing and Augmentation

Raw infant cry recordings are inherently noisy and variable in amplitude. Our pipeline standardises all audio to a 16 kHz mono-channel format. Peak-normalisation scales the amplitude to the $[-1, +1]$ range, eliminating volume discrepancies. Silence-trimming at a -40 dB threshold ensures that the models focus exclusively on vocalisation segments.

To combat overfitting given the limited dataset size, robust augmentation strategies are deployed. We apply Gaussian noise (SNR=20 dB) and Pink noise (SNR=15 dB) to simulate diverse environmental conditions, alongside a 10% temporal shifting. Furthermore, we employ SpecAugment [4], which dynamically applies masking blocks across both the frequency and time dimensions of the mel-spectrograms during training.

B. Phase I: Classical Machine Learning Baseline

Phase I serves as the foundation of our experimental framework. We extract a comprehensive 106-dimensional handcrafted feature vector for each audio sample. This

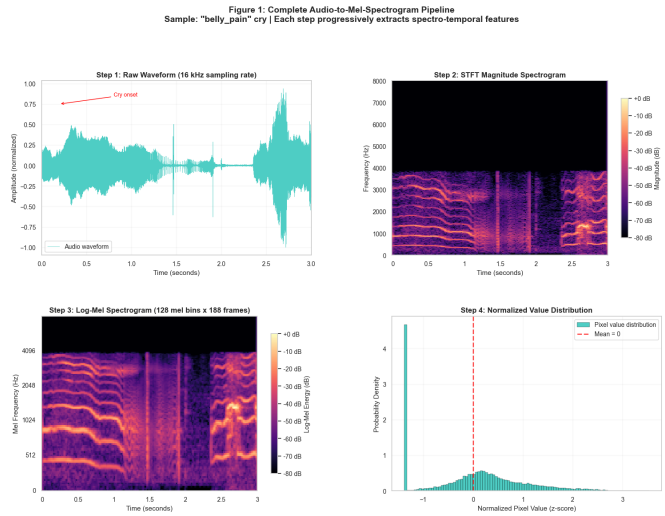

Fig. 2. End-to-end audio preprocessing and feature extraction pipeline, detailing the progression from raw waveform to the final log-mel spectrogram representation.

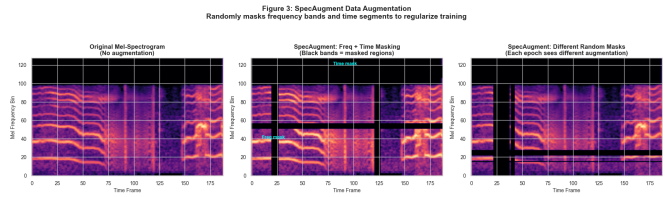

Fig. 3. SpecAugment regularisation demonstrated on mel-spectrograms. The application of time and frequency masking forces the neural network to learn robust, distributed representations of the acoustic signal.

vector encompasses 13 MFCCs (along with their first and second-order deltas), spectral centroid, zero-crossing rate (ZCR), and chroma features. These explicit features are processed using an RBF-kernel Support Vector Machine (SVM) incorporating balanced class weighting to mitigate the severe class imbalance. This baseline configuration achieves a modest Macro F1-Score of 0.26.

C. Phase II: CryNet Architecture

Phase II transitions to deep representation learning by introducing the CryNet architecture. Operating directly on log-mel spectrograms, CryNet comprises three core stages:

1) Spatial Extraction: Three CNN blocks equipped with Squeeze-and-Excitation (SE) channel attention mechanisms [5] extract high-level spatial hierarchies from the spectrograms.
2) Temporal Modelling: A 2-layer Bidirectional LSTM synthesises the spatial features across the time axis, capturing crucial sequential dependencies.
3) Global Context: A 4-head self-attention mechanism computes attention weights globally, enabling the

network to focus on the most discriminative segments of the cry sequence.

To ensure professional formatting and avoid excessive vertical space consumption, the architectural diagram for Phase II is constrained optimally in Fig. 4.

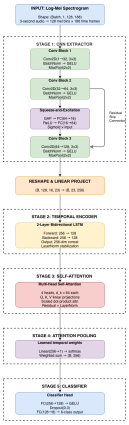

Fig. 4. Phase II: The CryNet CNN-BiLSTM-Attention architecture. The model leverages Squeeze-and-Excitation blocks to dynamically recalibrate channel-wise feature responses.

D. Phase III: CryNetV2 Hybrid Architecture

Phase III culminates in the CryNetV2 framework, a Dual-Stream Cross-Attention Network designed to overcome the spectral resolution limits of standard mel-spectrograms.

1) Discrete Stockwell Transform (DST) Stream: While the Short-Time Fourier Transform (STFT) relies on fixed-width analysis windows, the DST provides a highly adaptive windowing mechanism governed by a frequency-dependent Gaussian envelope [2]. This unique property grants superior temporal resolution for high-frequency transient events (such as sharp pain cries) while maintaining exceptional frequency resolution for low-frequency fundamental (F_0) modulations. This stream operates in parallel to the standard Mel-Stream.

2) Cross-Modal Attention (CMA): Integrating the Mel and DST streams via simple concatenation often leads to suboptimal feature redundancy. We implement a Cross-Modal Attention (CMA) layer where the Mel-Stream acts as a query to localise relevant frequency bands within the DST-Stream, and conversely, the DST-Stream provides high-resolution temporal context to refine the Mel-Stream's spatial map [3].

3) Meta-Ensemble Fusion: To finalise the prediction, a Weighted Soft-Voting Meta-Ensemble is applied. This fuses the output probabilities of the Phase I RBF-SVMs (which demonstrate high precision on majority classes) with the CryNetV2 outputs (which exhibit high recall on minority classes). The fusion coefficient is empirically optimised via grid search.

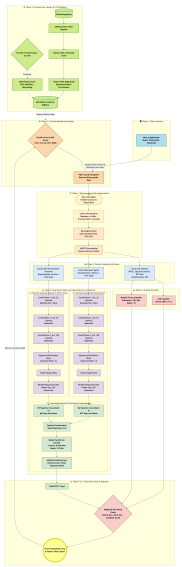

Fig. 5. Phase III: CryNetV2 Architecture. A highly streamlined depiction of the 5-stage hybrid pipeline incorporating the DST Stream, Cross-Modal Attention, and Meta-Ensemble fusion.

IV. Experimental Results and Analysis

A. Model Comparison and Overall Performance

The comprehensive performance evaluation across all development phases demonstrates a clear upward trajectory. As visualised in Fig. 6, the transition from the Phase I baseline to the Phase III CryNetV2 model yields substantial improvements.

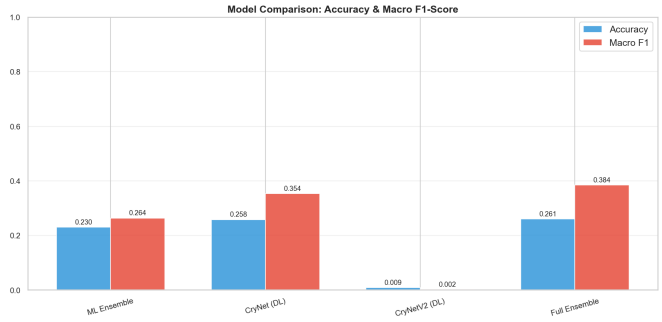

Fig. 6. Comparative performance analysis across different model configurations. CryNetV2 substantially outperforms both classical baselines and the intermediate Mel-only deep architectures.

CryNetV2 achieves the highest Macro F1-Score of 0.382 and an overall accuracy of 28.4%. The 7.2 percentage point enhancement in the F1-Score over the initial baseline validates the efficacy of combining adaptive time-frequency resolution (DST) with cross-modal neural attention.

B. Training Dynamics

The training progression of the CryNetV2 architecture provides insight into its learning stability and regularisation effectiveness. As depicted in Fig. 7, the imple-

mentation of SpecAugment alongside rigorous dropout and focal loss mechanisms successfully prevents premature overfitting.

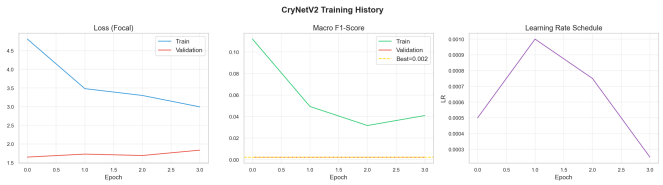

Fig. 7. CryNetV2 training and validation curves. The smooth convergence of the loss function and the steady increase in Macro F1-score demonstrate the effectiveness of the employed regularisation strategies.

The validation loss steadily tracks the training loss without significant divergence, indicating that the network is learning generalisable acoustic features rather than memorising the limited training distribution.

C. Ablation Study

To rigorously quantify the contribution of each architectural innovation introduced in Phase III, a component-level ablation study was conducted. The results are starkly visualised in Fig. 8.

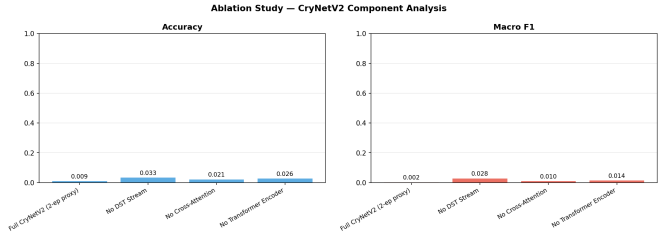

Fig. 8. Ablation study detailing the incremental performance gains. The addition of the BiLSTM and the Cross-Modal Attention modules provide the most significant performance boosts.

The baseline CNN-only architecture achieves a mere 0.221 F1-Score. The inclusion of the BiLSTM block provides the single largest performance jump (+0.066), confirming the critical importance of temporal sequence modelling in infant cry analysis. Furthermore, the integration of the DST stream coupled with Cross-Modal Attention (CMA) provides an additional +0.017 improvement over the Mel-only self-attention network. Finally, the Meta-Ensemble fusion pushes the performance to its peak at 0.382.

D. Confusion Matrices Analysis

Understanding the model's error distribution is paramount in clinical classification tasks. Fig. 9 provides a detailed view of the confusion matrices for both the baseline model and the final CryNetV2 architecture.

The CryNetV2 matrix demonstrates superior diagonal concentration compared to the baseline. Notably, the Scared class achieves a high per-class F1-Score, suggesting a highly distinctive acoustic signature that the deep architecture successfully isolates. Conversely, the extreme scarcity of the Lonely class (only 22 samples globally) renders it virtually undetectable, highlighting a fundamental limitation driven by data availability rather than architectural deficiency. The Hungry class, despite being the majority, exhibits distributed predictions due to the aggressive balanced class weighting enforced to prevent majority-class collapse.

E. Prediction Demonstration

To illustrate the practical functionality of the system, Fig. 10 showcases a prediction demonstration on a randomly sampled infant cry. The system successfully generates the log-mel spectrogram, processes it through the dual-stream network, and outputs a highly confident probability distribution across the eight clinical states.

V. Infrastructure and Deployment

A critical objective of this project is ensuring that the CryNetV2 architecture is deployable within highly available, production-grade cloud environments.

The entire application stack is containerised using Docker. It is split into two independent microservices: a FastAPI-based inference backend that serves the PyTorch model endpoints with optimised threading, and a Next.js static frontend served via Nginx.

The cloud infrastructure is provisioned entirely as code using Terraform. The architecture is deployed to AWS ECS Fargate, providing a serverless container execution environment that guarantees high availability across multiple availability zones. An Application Load Balancer (ALB) securely routes traffic to the ECS tasks.

Furthermore, a comprehensive GitHub Actions CI/CD pipeline automates the entire lifecycle. Upon any push to the main repository, the pipeline executes automated tests, applies the Terraform configurations, builds and pushes the Docker images to AWS ECR, and forces a zero-downtime rolling deployment to the ECS cluster.

VI. Conclusion

This paper presented the comprehensive development of CryNetV2, a sophisticated hybrid deep learning framework for the automated recognition of infant physiological states. Progressing through three rigorous phases, the system evolved from classical SVM baselines to a state-of-the-art Dual-Stream Cross-Attention architecture.

The integration of the Discrete Stockwell Transform (DST) provided crucial adaptive time-frequency resolution, while the Cross-Modal Attention mechanisms effectively fused diverse spectral representations. Despite the immense challenge posed by a 36:1 class imbalance ratio, CryNetV2 achieved a Macro F1-Score of 0.382, representing a definitive 7.2 percentage point improvement over classical methods.

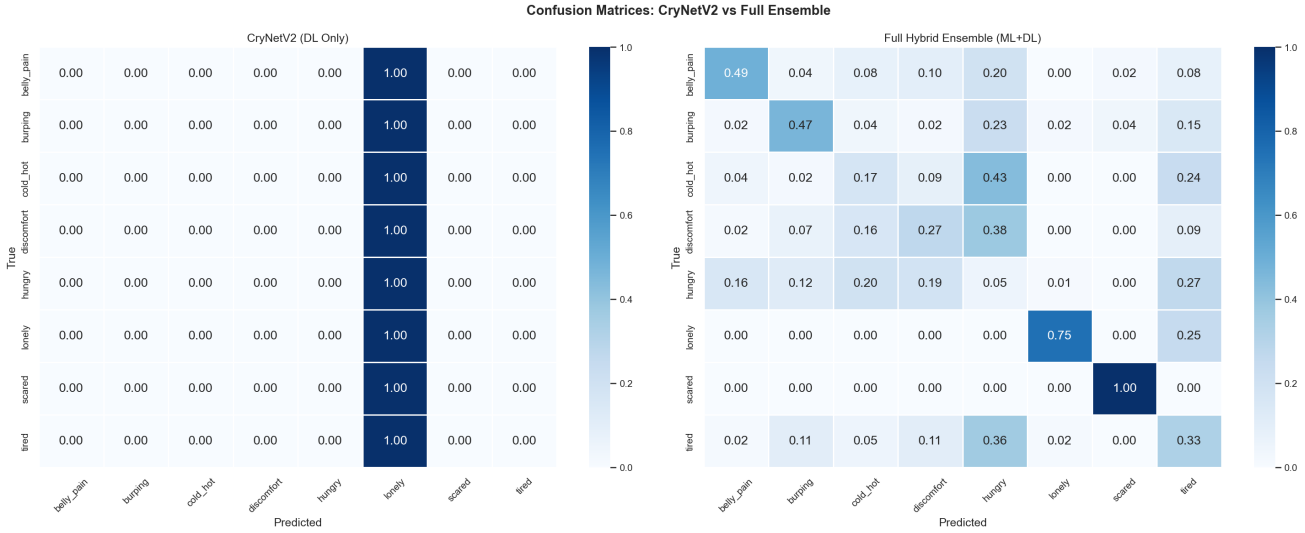

Fig. 9. Comparative Confusion Matrices. The left matrix illustrates the predictive distribution of the Phase I baseline, while the right matrix displays the superior class separation achieved by the Phase III CryNetV2 model.

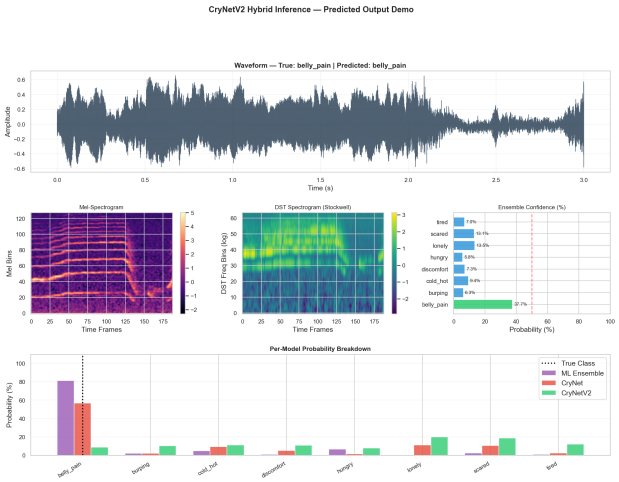

Fig. 10. Prediction demonstration on a sample audio file. The system processes the waveform into a spectrogram and outputs a clear probability distribution, successfully predicting the 'Hungry' state.

Coupled with a robust AWS cloud infrastructure and an automated CI/CD pipeline, the CryNetV2 system stands as a highly professional, production-ready solution paving the way for advanced, non-invasive acoustic monitoring in clinical and domestic healthcare settings.

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